GlucoTwin: A Digital Twin that Warns of Post-Meal Glucose Spikes 2 Hours Ahead

Happiest Health Digital Twin Challenge 2026

The Goal Diggers

Raman Kumar, Aditya Mehrda, Deepak Verma, Sahil Pazare

Indian Institute of Information Technology,
Design and Manufacturing, Kurnool

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Outline

1. Motivation
2. Data and architecture
3. Methods
4. Validation
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7. Conclusion

The problem in India

Scale

101 million

people living with diabetes

136 million

more with prediabetes, the early stage

ICMR-INDIAB, *Lancet Diabetes & Endocrinology*, 2023

- **Carbohydrate-heavy meals:** rice, roti and sweets drive post-meal spikes.
- **Personal responses:** the same meal raises glucose differently in two people.
- **A blind spot in the clinic:** doctors see HbA1c every few months, not what happens after lunch.
- Repeated post-meal spikes are linked to poorer long-term control.

Our idea: a virtual copy of one patient's glucose metabolism

GlucoTwin fuses a **static health record** with **dynamic wearable data** and:

Forecast	Warn	What-if	Explain
Glucose 30, 60, 90 and 120 minutes ahead, with an 80% uncertainty band.	Probability of a spike within 2 hours: glucose reaching 180 mg/dL or rising 50 mg/dL.	A smaller portion, millets instead of rice, a 15-minute walk: see the effect first.	Plain-language drivers of each forecast; personalized after the first 3 days.

Use case. A diabetologist opens the dashboard at a follow-up: the clinic overview ranks patients by current spike risk, the twin shows which meals push one patient above 180 mg/dL, and a simulated alternative (half the rice plus a walk) shows a lower peak, so the advice is personal, not generic.

Data: CGMacros (PhysioNet, open access)

45

adults, about 10 days each, free living

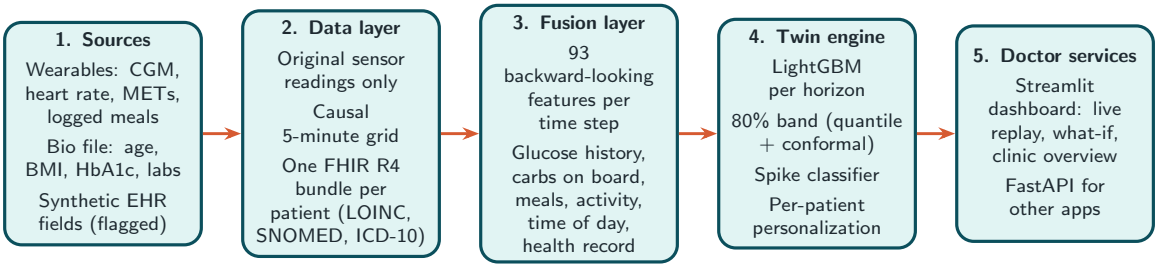
15 healthy
16 prediabetes
14 type 2 diabetes

License CC BY-NC-SA 4.0: the raw data is never committed to our repository.

Stream	Source	Content
Static: health record	CGMacros bio file	Age, sex, BMI, HbA1c, fasting glucose and insulin, lipid panel
Static: synthetic	Generated, flagged in every record	Diagnoses, medications, TCF7L2 genotype. Shown in the dashboard, **never used as model input**
Dynamic: wearables	CGMacros participant files	Dexcom G6 and Libre Pro glucose sensors, Fitbit heart rate and METs, logged meal macronutrients

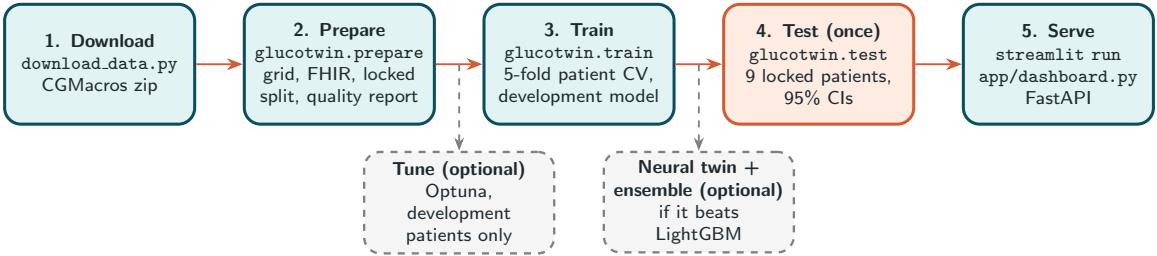
Key preprocessing step. CGMacros stores 1-minute glucose rows that were interpolated between real readings, so each value partly contains the *next* reading. GlucoTwin keeps only the original sensor readings and builds a strictly causal 5-minute grid.

System architecture



Read left to right: data become a causal, FHIR-backed record; features are fused; the engine forecasts, bands and alerts; doctors use the results through the dashboard or the API.

End-to-end workflow



Every step reads the same configuration file, so results are reproducible; the test step refuses models trained on a different dataset or split.

Health record as FHIR R4

Resource	Code	Meaning
Observation	LOINC 4548-4	HbA1c
Observation	LOINC 1558-6	Fasting glucose
Observation	LOINC 20448-7	Fasting insulin
Observation	LOINC 39156-5	Body mass index
Condition	SNOMED 44054006, ICD-10 E11.9	Type 2 diabetes
Condition	SNOMED 714628002, ICD-10 R73.03	Prediabetes
MedicationStatement	flagged as synthetic	Medications

- One bundle per patient.
- FHIR R4 is the format used by India's Ayushman Bharat Digital Mission (ABDM).
- The model reads its static features back *from this record*, not from a spreadsheet.
- Diagnosis follows baseline HbA1c with ADA thresholds: 5.7% and 6.5%.

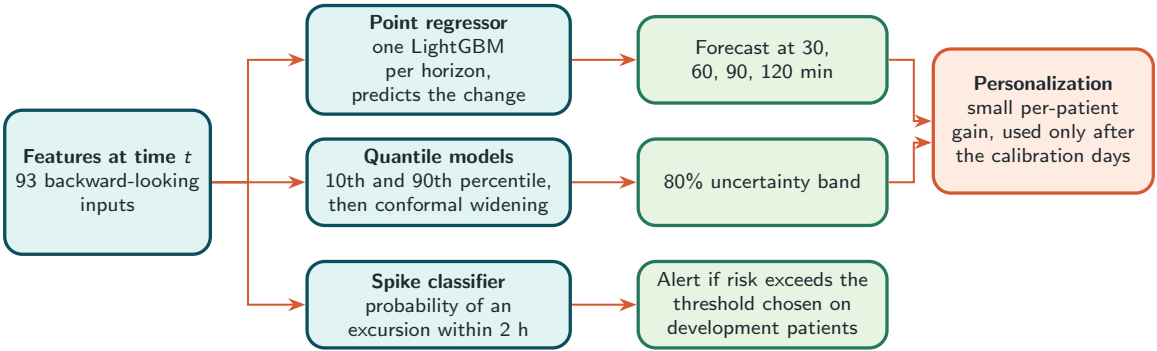
Fusion layer: 93 features, all looking backwards

Feature group	What it captures
Glucose history	Level, trend, variability and 6-hour context
Carbs on board	Gamma absorption curve from logged meals
Meal macros	Carbohydrate, protein, fat and fibre by time window
Activity	METs and heart rate relative to rest
Time of day	Daily rhythm of glucose control
Missing-data flags	Sensor gaps stay visible and are never filled in
Health record	HbA1c, HOMA-IR, TG/HDL, BMI, age, diagnosis
Plan inputs	Planned meal and walk; used *only* by the what-if tool and hidden in every reported forecast

Leakage guard

A test proves that changing raw data after time T never changes any model input at or before T.

Twin engine: how one forecast is produced



Design choices that matter

Physiology built in

Monotone constraints: more carbohydrate never lowers a forecast or a risk; more planned activity never raises it.

Personalization

A per-patient correction learned from the first 3 days, kept for a horizon only if it lowers error on a purged check.

Honest uncertainty

Quantile models plus conformal calibration on validation patients, so the 80% band really holds the outcome about 80% of the time.

Alert threshold

Chosen on development patients so that at most 10% of non-spike times raise an alert.

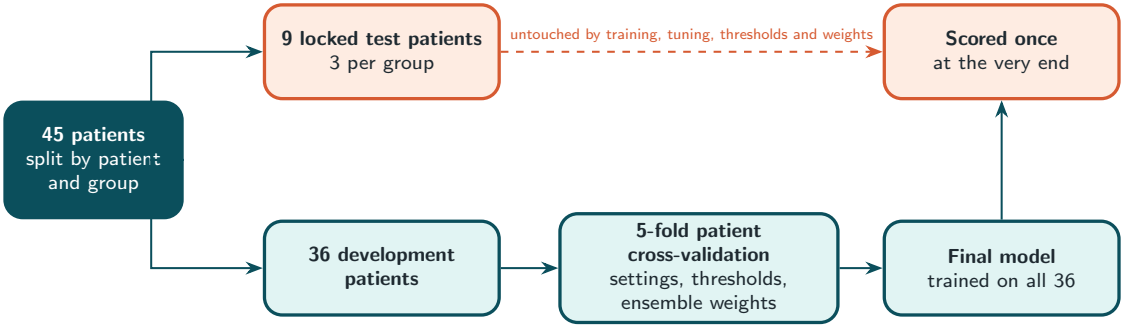
Plan dropout

The model is trained with and without knowing future meals, so reported forecasts are made without a plan and the what-if tool can still use one.

Why trees, not a neural network

We also built a neural twin (P-MRGF: causal TCN with FiLM conditioning). With 45 patients it was less accurate (30.1 vs 27.4 mg/dL at 2 hours on the test patients), so LightGBM is the deployed engine.

Validation design: nothing about a test patient is ever used



Inside every scored patient

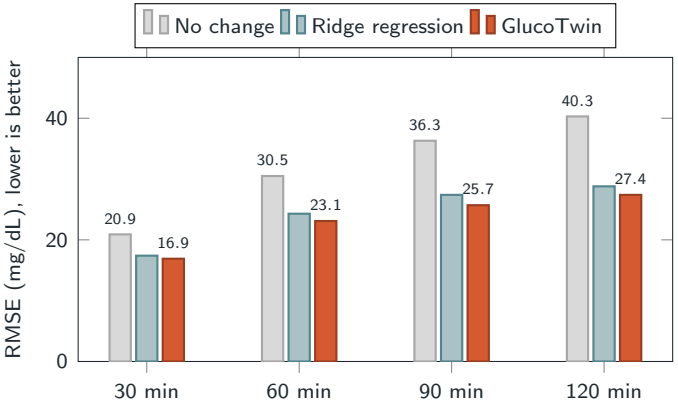


Baselines are scored on the same rows: no change, linear trend, ridge regression.

How the pipeline prevents leakage

Way a model can cheat	**Safeguard (each one has a test)**
Interpolated 1-minute glucose contains the next reading	Keep only original readings; right-labelled 5-minute grid; last reading carried forward at most 10 minutes
A feature uses future values	Every feature is a backward-looking window
Same person in training and test	All splits are by patient, never by row or window
Choosing settings on the reported numbers	Locked 20% test set excluded from tuning, cross-validation, thresholds and ensemble weights
Calibration overlaps scored days	Purge gap of one full horizon between calibration and scoring
Testing models trained on other data	A data fingerprint is checked; the test step refuses mismatches
Peeking at the test set repeatedly	Every test run is logged; the test was run once

Forecast accuracy on the 9 locked test patients



Locked test set, run once. Development cross-validation (36 patients): 17.4, 24.4, 28.0 and 30.1 mg/dL.

27.4 mg/dL

error 2 hours ahead

(95% CI 22.5 to 32.2)

32% lower

than assuming glucose stays the same

(40.3 mg/dL)

98.9% +

of forecasts in Clarke zones A+B at every horizon

Forecast error in detail (RMSE, mg/dL)

Model (locked test patients)	**30 min**	**60 min**	**90 min**	**120 min**
Persistence (no change)	20.9	30.5	36.3	40.3
Linear trend (last 30 min)	28.5	50.3	65.4	76.6
Ridge regression	17.4	24.3	27.4	28.8
GlucoTwin (LightGBM)	**16.9**	**23.1**	**25.7**	**27.4**
95% CI (bootstrap over patients)	14.9–18.6	20.2–26.0	21.5–29.9	22.5–32.2
GlucoTwin personalized	16.5	22.8	25.2	26.6
same rows, population model	16.9	23.1	25.5	27.0
Oracle: real future meals given (not a forecast)	16.7	22.9	25.5	26.7

- GlucoTwin lowers the error of "no change" by 19% at 30 min and 32% at 120 min.
- The gap to ridge regression is smaller than the confidence intervals, so we do not overclaim it.
- Personalization adds a further 0.3 to 0.4 mg/dL after 3 days.

Clinical accuracy (Clarke error grid, locked test patients)

Horizon	MAE	MARD	Zone A	Zone A+B	Zone D+E	80% band
30 min	11.5	8.5%	90.8%	99.7%	0.34%	79.8%
60 min	16.2	12.0%	81.7%	99.4%	0.53%	78.9%
90 min	18.3	13.5%	76.7%	99.2%	0.72%	78.5%
120 min	19.7	14.4%	73.8%	98.9%	0.99%	79.4%

- MAE in mg/dL; MARD is the mean absolute relative difference.
- Zones A and B are clinically acceptable; zones D and E would lead to a wrong treatment decision (about 1% at 2 hours).
- The 80% band contains the real value 78.5 to 79.8% of the time, so the band a doctor sees is honest.

Early warning of post-meal spikes

0.85

AUROC (95% CI 0.80 to 0.89)

96%

of 160 crossings warned in advance, median warning 2 h

2.7

false alerts per patient-day: the price of catching them

At the development threshold	Locked test	Dev. CV
Sensitivity / specificity	0.67 / 0.84	0.37 / 0.94
Precision	0.45	0.52
Crossings warned in advance	96%	67%
False alerts per patient-day	2.66	1.62
Share of time an alert shows	32%	19%

Test rows: 22,262 (15.8% positive); AUPRC 0.550; Brier score 0.128. Accuracy is deliberately not reported: with 16% positives it rewards a model that never alerts.

Honest reading

The ranking of risk is stable, but the threshold, set on a few development patients, alerted more often on the test patients. Adapting it to each patient during the calibration days is our next step.

What each data stream adds (development cross-validation, 36 patients)

Inputs (RMSE, mg/dL)	30 min	60 min	90 min	120 min	Spike AUROC
CGM only	18.3	25.7	29.4	31.2	0.774
+ health record	18.3	25.6	29.2	31.0	0.770
+ meals	17.4	24.5	28.1	30.1	0.802
+ activity	17.5	24.6	28.1	30.1	0.800
GlucoTwin, all inputs	**17.4**	**24.4**	**28.0**	**30.1**	**0.799**
All but health record	17.2	24.4	28.2	30.3	0.793

- **Meals** give the largest gain: about 1 mg/dL less error at every horizon.
- The **health record** helps a little, mostly at 90 to 120 minutes.
- **Wrist activity and heart rate** carried little signal in this mostly sedentary cohort.
- Differences of 0.1 to 0.2 mg/dL are within noise; the full model was kept, as decided before the test.

Models compared on the locked test patients

Model (RMSE, mg/dL)	30 min	60 min	90 min	120 min	Spike AUROC
No change (persistence)	20.9	30.5	36.3	40.3	–
Linear trend, last 30 min	28.5	50.3	65.4	76.6	–
Ridge regression	17.4	24.3	27.4	28.8	–
P-MRGF neural twin	18.0	25.5	28.8	30.1	0.794
Ensemble, LightGBM + P-MRGF	16.9	23.0	25.8	27.4	0.848
GlucoTwin, LightGBM (deployed)	**16.9**	**23.1**	**25.7**	**27.4**	**0.850**
Oracle: real future meals given	16.7	22.9	25.5	26.7	–

- The ensemble only matched LightGBM, so the simpler model is deployed.
- The oracle row is not a forecast: knowing the real future meals and activity lowers 2-hour error only from 27.4 to 26.7 mg/dL.

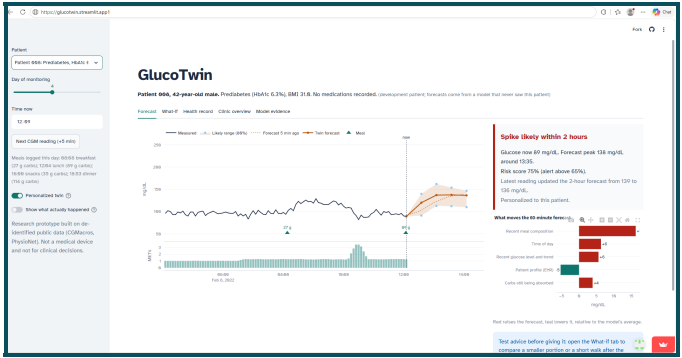
Results by metabolic group and R^2 (locked test patients)

Group (3 patients each)	30 min	60 min	90 min	120 min	Spike AUROC
Healthy	15.2	18.1	19.2	19.9	0.787
Prediabetes	17.1	22.4	24.0	25.5	0.830
Type 2 diabetes	18.2	27.7	32.2	34.7	0.801

R^2	30 min	60 min	90 min	120 min
Of the predicted change (honest measure)	0.345	0.426	0.498	0.536
Absolute glucose, GlucoTwin	0.769	0.570	0.465	0.393
Absolute glucose, no change	0.648	0.251	−0.064	−0.309

- Errors are largest for type 2 diabetes, where accurate forecasts matter most; 3 patients per group, so group results are indicative only.
- R^2 of absolute glucose flatters any model because "no change" already scores 0.648 at 30 minutes.

Doctor dashboard



Forecast. Live replay, 2-hour forecast, alert and drivers.

What-if. Build an Indian meal, add a walk, compare peaks.

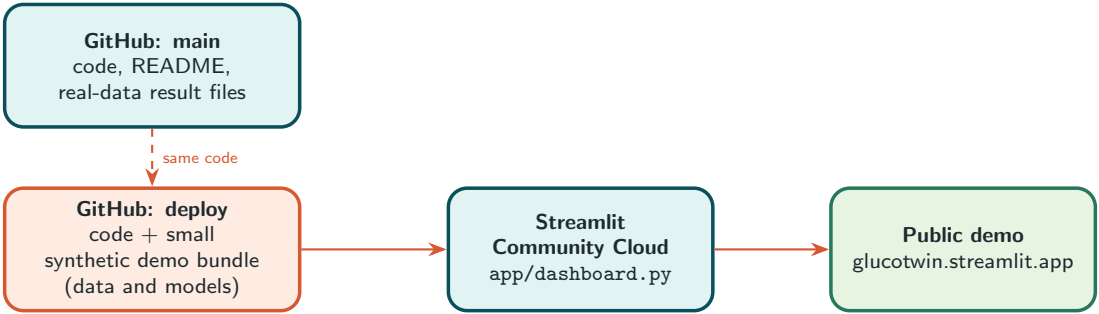
Health record. The FHIR bundle exactly as the twin reads it.

Clinic overview. Every patient ranked by spike risk.

Model evidence. Locked-test results and figures.

Live app: glucotwin.streamlit.app

Deployment



Transparency

The hosted demo runs on **synthetic data**, because the CGMacros license does not allow redistribution. Every number in this presentation comes from the real CGMacros data and the locked test patients.

Limitations and road to real use

Limitations

- 45 people in the USA, 10 days each.
- Meals are self-reported; unlogged snacks look like unexplained rises.
- Alert threshold set on a few patients: an alert showed 32% of the time on test.
- Wrist activity carried little signal, so a walk changes the forecast only slightly.
- Indian meal values are approximate until checked against IFCT 2017.

Road to real use

- Validation study with Indian patients and Indian meals.
- Alert thresholds adapted to each patient during calibration.
- Records exchanged as FHIR through ABDM.
- CDSCO approval as software as a medical device.
- Consent and data handling under the DPDP Act 2023.

Research prototype for a student competition. Not a medical device and not for clinical decisions.

Summary

- **Twin:** fuses a FHIR health record with wearable data into a forecast, an 80% band, a spike alert and a what-if simulator.
- **Accuracy:** 27.4 mg/dL error 2 hours ahead on unseen patients, 32% lower than assuming no change; 99% of forecasts in Clarke zones A+B.
- **Early warning:** AUROC 0.85, 96% of crossings warned in advance, at a cost of 2.7 false alerts per patient-day.
- **Rigour:** patient-level splits, 9 locked test patients scored once, leakage tests; negative results (neural network, ensemble) reported honestly.
- **Next:** per-patient alert thresholds and validation on Indian patients and meals.

Code: github.com/rmnkumrydv-cloud/GlucoTwin Live demo: glucotwin.streamlit.app

References

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Thank you

The Goal Diggers, IIITDM Kurnool

github.com/rmnkumrydv-cloud/GlucoTwin glucotwin.streamlit.app